CHAPTER

Functionalized biopolymer-based coatings and adhesives

15

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15.1 Introduction

The polymerization of biological molecules under natural conditions, especially during their growth cycle, can lead to biopolymer creation. This type of natural polymer represents excellent features, including flexibility, tensile strength, steadiness, reusability, and other characteristics. Biopolymers can be classified into nonsynthetic and synthetic groups (Table 15.1). Nonsynthetic biopolymers are derived from abundant and sustainable resources, such as starch, cellulose, chitin, and proteins. Due to the biodegradability of nonsynthetic biopolymers, they can be decomposed by microorganisms in natural environments, reducing waste accumulation and pollution. However, nonsynthetic biopolymers demonstrate some challenges and limitations, including low mechanical strength, poor thermal stability, and increased water absorption, which restrict their use in some applications. Nonsynthetic biopolymers may require special processing techniques, such as solvent casting, electrospinning, and freeze-drying, which are more complex and expensive than conventional techniques. On the other hand, synthetic biopolymers are polymers derived from renewable resources or can be biodegraded by microorganisms (Mtibe et al., 2021).

In recent years, considerable attention has been focused on biopolymers as conventional petroleum-based polymers to reduce serious environmental problems. Some of the most common synthetic biopolymers include polylactide (PLA), poly(ε-caprolactone) (PCL), poly(butylene succinate) (PBS), poly(butylene-adipate-*co*-terephthalate) (PBAT), and poly(hydroxyalkanoates) (PHAs). These polymers can be synthesized by different methods, such as polycondensation, ring-opening polymerization, and microbial fermentation. They can also be blended or reinforced with other biopolymers or nanomaterials to improve their properties and performance (Ayu et al., 2018; Muller et al., 2017; Reddy et al., 2013; Romani et al., 2017; Siracusa et al., 2015; Zafar et al., 2016). The enhanced features of synthetic biopolymers and their biocompatibility and functionality made them suitable materials for use in various sectors, such as food packaging, biomedical, agricultural, and textile. For example, PLA is widely used for thermoformed

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Table 15.1 Comparison of natural and synthetic biopolymers based on various characteristics.

Characteristic	Natural biopolymers	Synthetic biopolymers
Source	Derived from living organisms.	Man-made or synthesized in laboratories.
Composition	Proteins, nucleic acids, lipids, and carbohydrates.	Synthetic polymers inspired by nature.
Biodegradability	Varies; generally biodegradable.	Biodegradable, but rates can vary.
Renewable source	Yes.	Depends on the specific synthetic polymer.
Structural diversity	Highly diverse and complex structures.	Tailored for specific applications.
Production process	Often involves biological processes.	Typically involves chemical synthesis.
Functionalization	Natural functionalities.	Tailored for specific functions through chemical modifications.
Applications	Food industry, pharmaceuticals, textiles and clothing, and biomedical applications.	Biodegradable plastics, medical devices and implants, controlled drug delivery systems, and tissue engineering.
Challenges	Limited control over properties of natural biopolymers.	Balancing biodegradability with durability and functionality.

products, such as trays, cups, and bottles, due to its good transparency, stiffness, and biodegradability. PCL is suitable for drug delivery systems, tissue engineering scaffolds, and wound dressing materials owing to its low melting point, high flexibility, and biocompatibility. Because of their ductility, toughness, and composability, PBS and PBAT are leading materials for flexible packaging, such as films, bags, and containers. PHAs are versatile polymers that can be tailored to different applications, such as mulch films, disposable cutlery, and medical implants, depending on their chemical structure, molecular weight, and crystallinity (Mtibe et al., 2021).

Despite the resilient features of biopolymers, like other materials, they also face some challenges and limitations, such as high cost, low production yield, poor processability, high hydrophilicity, low thermal stability, and brittleness. Therefore further research and development are needed to overcome these drawbacks and to optimize the synthesis, processing, and characterization of synthetic biopolymers. Moreover, the environmental impact of synthetic biopolymers should be assessed by using lifecycle analysis and considering end-of-life scenarios, such as recycling, composting, biodegradation, and incineration. By addressing these issues, biopolymers can be more competitive and sustainable in the polymer market and contribute to developing a circular bioeconomy.

While biopolymers have gained significant attention for their potential to mitigate various concerns, enhancing their functionalities through strategic functionalization has emerged as a crucial area of exploration. Functionalization involves modifying the chemical structure of biopolymers to impart specific properties, addressing their inherent limitations (Hedar et al., 2023). Various methods, including grafting, blending, and copolymerization, enable the incorporation of functional groups, leading to improvements in mechanical strength, thermal stability, and hydrophobicity (Atz Dick et al., 2017; Fredi & Dorigato, 2023; Rajeswari et al., 2021).

Adhesives and coatings play pivotal roles across industries, from enhancing material durability to facilitating intricate bonding processes. Integrating functionalized biopolymers into adhesive and coating formulations introduces a new dimension of performance and sustainability. The tailored properties achieved through functionalization contribute to adhesive strength, durability, and the ability to adhere to various substrates. Simultaneously, coatings benefit from improved resistance to environmental factors, such as moisture, UV radiation, and temperature fluctuations (Nathanael & Oh, 2020; Yu & Cha, 2023).

The primary objective of this chapter is to provide a comprehensive understanding of the utilization of functionalized biopolymers in the field of adhesives and coatings. By addressing the challenges associated with both nonsynthetic and synthetic biopolymers, we aim to highlight the transformative potential of functionalization in overcoming limitations and expanding the application scope of biopolymer-based materials. Furthermore, the chapter seeks to elucidate the role of functionalized biopolymers in fostering sustainable practices within industries, offering insights into developing eco-friendly coatings and adhesives that align with the principles of a circular bioeconomy. Through a detailed exploration of recent advancements and case studies, this chapter aims to inspire further research and innovation in the field, paving the way for enhanced biopolymer-based materials with tailored functionalities.

15.2 Functionalization of biopolymers

The enhanced performance and expanded applicability of biopolymers in various applications often require their structural modification and, more specifically, their surface functionalization. In surface modification, along with introducing new functional groups from molecules or nanoparticles, the properties of whole polymeric molecules, such as biocompatibility, bioactivity, hydrophilicity, hydrophobicity, adhesion, conductivity, optical, magnetic, or catalytic properties, can be improved. Surface functionalization can also enable the immobilization of biomolecules, such as enzymes, antibodies, DNA, or drugs, on biopolymer surfaces for biosensing, bioimaging, biocatalysis, or drug delivery. Biopolymer surface functionalization is a complex process that requires careful consideration of the desired properties and the methods used to achieve them. However, it is an important process with many applications in developing new materials and technologies. Different methods have been employed to modify the surface of biopolymers. The type of biopolymers, the desired functionality, and the application are critical parameters to select the best modification methods. Some of the regular methods used for surface modification of biopolymers, such as blending, chemical linkages, cross-linking, grafting, and bio- or nanocomposite formation (Fig. 15.1) (Igarzabal et al., 2015; Mtibe et al., 2021). Modifying the surface by introducing functional groups can alter the hydrophilic/hydrophobic balance, increase the mechanical strength and chemical stability, control the biodegradability and solubility, and incorporate active substances or fillers into the biopolymer matrix (Table 15.2).

15.2.1 Importance of functionalization

Functionalization plays a pivotal role in enhancing the inherent properties of biopolymers, addressing their limitations, and expanding their range of applications. Chemical modifications can bolster mechanical strength, chemical stability, and resistance to environmental factors. On the other hand, physical modifications offer avenues for fine-tuning surface characteristics without compromising the bulk

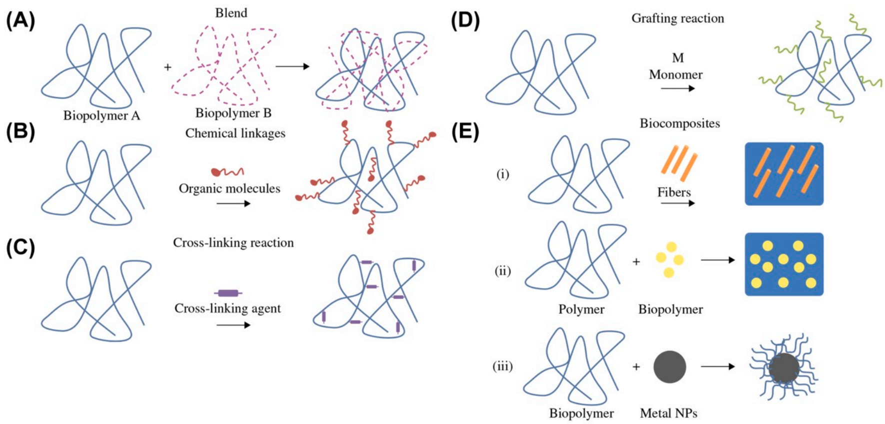

FIGURE 15.1 Innovative strategies for transforming biopolymers.

(A) Blends, (B) chemical linkages, (C) cross-linking, (D) grafting, and (E) biocomposite formation (i, biopolymer matrix; ii, biopolymer fillers; and iii, surface modified inorganic materials).

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Table 15.2 Commonly used approaches in surface modification of biopolymers and their advantages and disadvantages.

Method	Description	Advantages	Disadvantages	Examples	Refs.
Blending	Mixing biopolymers with other polymers or additives to form new materials with improved properties	Versatile and simple, can enhance compatibility, mechanical strength, solubility, and biodegradability	May affect the bioactivity, purity, and stability of biopolymers	Chitosan–alginate, starch–polyethylene, and pHEMA–chitosan	Fredi and Dorigato (2023)
Chemical linkages	Introducing small functional groups to the structure of biopolymers, such as alkyl, polar, or dendritic groups	Can change hydrophilic/ hydrophobic balance, increase solubility, and impart specific properties	May require harsh reaction conditions, affect biocompatibility, and introduce toxicity	Carboxylation, dendronization, and functionalization with molecules, such as hyaluronic acid	Aldana et al. (2012), Masteri-Farahani and Shahsavarifar (2021)
Cross-linking	Forming covalent bonds between biopolymer chains using different chemical reagents or physical stimuli	Can enhance mechanical strength, chemical stability, control aqueous permeability and solubility, and decrease swelling	May affect the biodegradability, bioactivity, and toxicity of biopolymers, depending on the cross-linking agent and degree	Glutaraldehyde-cross-linked chitosan, genipin cross-linked chitosan, alginate–polyvalent cations	Sun, Liu, et al. (2020)
Grafting	Attaching synthetic monomers or polymers onto biopolymer chains via radical or other initiation methods	Can introduce desired properties of synthetic polymers, such as thermal and physical stability, optimal swelling, and solubility	May affect the biodegradability, bioactivity, and toxicity of biopolymers, depending on the grafting agent and degree	Methylmethacrylate-grafted chitosan, polyethylene-grafted starch, and poly(*N*-vinyl-2-pyrrolidone)-grafted chitosan	Datta and Genzer (2016), Henze et al. (2014)
Bio- or nanocomposites	Incorporating reinforcing fillers, such as natural fibers, nanoclays, or metal nanoparticles, into biopolymer matrices	Can improve the mechanical, thermal, barrier, optical, and electrical properties of biopolymers	May affect the biodegradability, bioactivity, and toxicity of biopolymers, depending on the filler type and amount	Starch–cellulose fibers, alginate–nanoclay, and chitosan–silver nanoparticles	Madhi and Hadavand (2022), Mondal and Purkait (2021)

properties, resulting in biopolymers with improved performance across various domains. Tailoring the functionality of biopolymers through surface modification opens new possibilities for their utilization. By selectively introducing functional groups or nanoparticles, biopolymers can be customized for specific applications, ranging from biomedical devices to packaging materials. The controlled adjustment of hydrophilicity, adhesion properties, and biodegradability empowers researchers and industries to design materials that meet precise requirements. One of the notable outcomes of biopolymer functionalization is the ability to achieve controlled release of active substances. Whether for drug delivery or the gradual release of additives in coatings, functionalization techniques allow for incorporating substances with controlled and sustained release profiles. This feature is particularly significant in biomedical applications, where drug-controlled release can improve therapeutic outcomes and minimize side effects (Duceac & Coseri, 2022; Platnieks et al., 2023; Singh et al., 2023).

15.2.2 Types of functionalization

15.2.2.1 Chemical functionalization techniques

Chemical functionalization stands out as a versatile approach for modifying biopolymer surfaces. This technique forms covalent bonds between the biopolymer and functional groups, leading to precise control over surface properties. Standard chemical functionalization methods include acylation, alkylation, esterification, and amidation. These methods enable the introduction of various chemical moieties, allowing for adjustments in hydrophilicity, chemical stability, and reactivity. Chemical functionalization provides a robust platform for tailoring biopolymers to specific applications, such as enhancing adhesion strength or facilitating controlled drug release (Ganie et al., 2020).

15.2.2.2 Physical functionalization techniques

Physical functionalization techniques focus on altering the surface properties of biopolymers without involving chemical reactions. These methods include plasma treatment, irradiation, and physical vapor deposition. Plasma treatment, for instance, enhances surface wettability and facilitates subsequent adhesion processes without compromising the bulk properties of the biopolymer. Physical functionalization techniques offer advantages, such as simplicity and minimal alteration of the biopolymer's chemical structure, making them suitable for applications where maintaining inherent properties is crucial (Ganie et al., 2020).

15.2.2.3 Enzymatic functionalization

Enzymatic functionalization presents a biocompatible and environmentally friendly alternative for modifying biopolymer surfaces. Enzymes catalyze specific reactions, enabling the introduction of functional groups with high precision. This technique is particularly valuable for applications in biomolecule immobilization, where enzymes can be strategically attached to enhance biocatalysis, biosensing, or drug delivery. Enzymatic functionalization showcases the potential for sustainable and controlled modifications of biopolymers, aligning with the principles of green chemistry (Aracri et al., 2014; Francesko et al., 2015). A study demonstrated the enzymatic functionalization of cork surface with antimicrobial hybrid biopolymer/silver nanoparticles. The team used laccase to catalyze the covalent immobilization of chitosan or aminocellulose-doped AgNPs onto cork matrices, aiming to enhance their antibacterial properties for water remediation applications. The study showed that the enzymatic grafting approach allowed for the successful modification of the material with stable and durable antibacterial coatings, which could be potentially used as filters in constructed wetlands (Francesko et al., 2015).

A comprehensive exploration of the described functionalization methods, encompassing both chemical and physical approaches, will be provided in subsequent sections. Specific attention will be devoted to elucidating diverse techniques employed in modifying biopolymers for adhesive or coating applications. The forthcoming sections will furnish a detailed account of these functionalization strategies, offering in-depth insights and concrete examples of their practical and potential implementation in adhesive and coating technologies.

15.3 Applications of functionalized biopolymers in coatings

The tunable physicochemical properties of functionalized biopolymers, biocompatibility, biodegradability, and bioactivity make them attractive coating materials because they can enhance the performance and functionality of biomaterials and devices. Different methods regarding the selected material type can be applied to create purpose-specific polymer coatings. The tendency of the polymers toward water can challenge their applications. Low hydrophobicity and limited surface area can dimmish the bioactivity and confine their incorporation into composites or coating implants. To overcome this lack, polymer integration with nanoparticles, owing to the benefits of both hardness adjustment and component distribution, has been proposed as a practical solution. Biodegradable polymer coatings on metal surfaces induce properties with substantial potential for diverse industrial applications, modifying qualities like elasticity, wettability, bioactivity, and adhesiveness (Reggente et al., 2017). Polymers, such as poly(2-methyl-2-oxazoline) with low molecular weight and moderate hydrolysis (Gnedenkov, Sinebryukhov, Mashtalyar, et al., 2014) and polytetrafluoroethylene with protective and antifriction features (Gnedenkov, Sinebryukhov, Zavidnaya, et al., 2014) demonstrate the versatility of polymer coatings. These advancements and many similar improvements highlight the multifaceted role of polymer coatings in enhancing the performance and biocompatibility of various materials in diverse applications.

The preparation of biopolymer-based coatings can be accomplished through various techniques, either physical or chemical in nature. In physical coating approaches, the biopolymer is directly deposited onto the substrate using specific technical processes, such as spin coating, dip coating, electrospinning, and vapor transport. On the other hand, chemical coating strategies involve chemical reactions between the biopolymer and the substrate. These methods often necessitate (at times intricate) chemical pretreatments of the biopolymer or the substrate surface. The selection of a specific coating approach for a given application is contingent on several factors. These include the chemical properties of the substrate, the desired coating thickness, properties, and economic considerations. The decision-making process involves evaluating the application's unique requirements and choosing the most suitable method accordingly. Whether to opt for a physical or chemical coating approach depends on the specific characteristics and needs of the substrate, aiming to achieve the desired coating results effectively and efficiently. The following subsections will introduce an overview of the most commonly used Physical and chemical coating approaches.

15.3.1 Physical coating approaches

Physical coating approaches encompass a spectrum of techniques that, despite their simplicity, offer versatility in the application of coatings onto various substrates. These methods are particularly suitable for scenarios where the coated material is not subjected to intense mechanical stress or abrasion. Physical coating methods fall into two main categories: dry and wet. Dry coating involves using polymer powders that are heated and sprayed onto the substrate. However, this technique faces challenges when applied to biopolymers, known for their thermal sensitivity. This method is incompatible with most biopolymers due to their susceptibility to thermal stress. Despite this limitation, dry coating remains widely utilized in industries where quick and even distribution of coatings is essential. Wet coating, utilizing polymer solutions, offers a more versatile approach and is applicable to biopolymers. Techniques, such as electrospinning, dip coating, and spin coating fall under this category and are crucial in tailoring coatings for specific applications (Fig. 15.2).

Layer-by-Layer (LbL) assembly represents an elegant and precise approach to create coatings with controlled thickness and composition. This technique involves the sequential deposition of alternating layers of oppositely charged polymers or other materials onto the substrate (Jeon et al., 2015).

Spray coating techniques involve the dispersion of polymer solutions or powders onto a substrate through the use of a spray gun. While this method is widely employed in various industries, it may pose challenges when applied to biopolymers, which are often sensitive to thermal stress. Adjusting parameters, such as spray pressure, nozzle size, and drying conditions becomes crucial in achieving uniform and effective coatings. Spray coating finds applications in areas where a quick and even distribution of coatings is essential (Mujtaba et al., 2022).

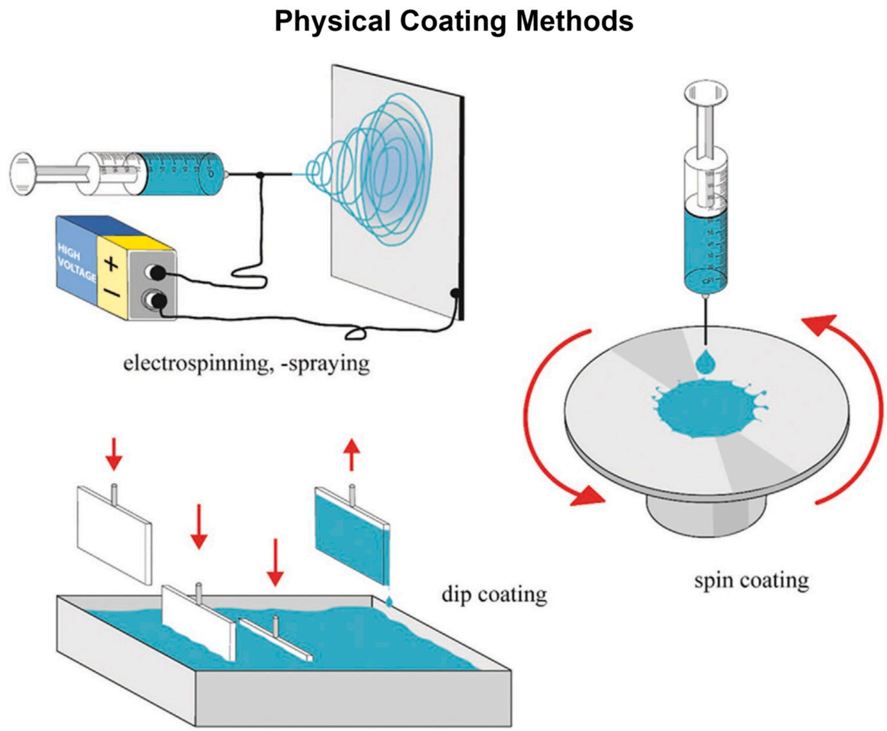

FIGURE 15.2 Varied techniques for depositing biopolymers onto substrates.

This includes electrospinning, dip coating, and spin coating.

Reused from Song, J., Winkeljann, B., & Lieleg, O. (2020). Biopolymer-based coatings: Promising strategies to improve the biocompatibility and functionality of materials used in biomedical engineering. Advanced Materials Interfaces, 7*(17). https://doi.org/10.1002/admi.202000850. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.*

Electrospinning stands out as a cutting-edge technique involving the application of an electric field to a polymer solution to create nanofibrous coatings deposited over a substrate. This method, known for its precision in controlling morphology and thickness, finds applications in diverse fields. Examples include tissue engineering, where the high surface area of electrospun coatings aids in cellular adhesion, filtration for efficient particle capture, and controlled drug release systems benefiting from tunable porosity (Akhouy et al., 2023; Greiner & Wendorff, 2007; Yang et al., 2018; Yarin et al., 2001).

Dip coating immerses the substrate into a polymer solution and then withdraws it at a controlled speed, allowing for the formation of uniform coatings. This technique is particularly effective in applications where a precise and even coating is crucial. Adjusting withdrawal speed and solution parameters provides control over coating thickness and morphology (Kian-Pour, 2023; Wu et al., 2016).

Spin coating involves spreading a solution evenly over the substrate through high-speed rotation. This method allows for fine-tuning coating properties by adjusting solution concentration, viscosity, and solvent evaporation rate. While effective, it does require substantial solution volumes and may result in material wastage (Kildeeva et al., 2023; Moreira et al., 2021; Spirk et al., 2021).

Other applications demanding ultrathin films (1–100 nm), advanced physical deposition methods like Langmuir–Blodgett, molecular self-assembly, LbL electrostatic deposition, and nanopatterning offer innovative solutions. These techniques, contributing to the development of advanced coatings, allow the creation of monomolecular or nanoscale layers of biopolymers with unique properties (Fujimori, 2015; Jeon et al., 2015; Lee et al., 2002).

15.3.2 Chemical coating approaches

Chemical coating methods are more complex and specific, but they offer the advantage of creating covalent bonds between the biopolymer and the substrate, which enhances the stability and durability of the coating. Chemical coating methods can be divided into "grafting to" and "grafting from." The former (grafting to) involves attaching the whole biopolymer molecule to the substrate by creating reactive groups on either or both. Grafting from involves attaching small initiator molecules to the substrate and then polymerizing the biopolymer directly on the surface (Fig. 15.3). Grafting to is more applicable for biopolymers, as most of them cannot be synthesized in situ. Depending on the functional groups present in the biopolymer, different coupling reactions can be used to create covalent bonds with the substrate. The most commonly used crosslinkers in coupling reactions are glutaraldehyde (Habeeb & Hiramoto, 1968), (3-aminopropyl)triethoxysilane (Facci et al., 2002), carbodiimide by using 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) and *N*-hydroxysuccinimide (NHS) couples (Staros et al., 1986), and click chemistry (Gacal et al., 2006).

15.3.2.1 Crosslinking strategies

Glutaraldehyde is a crosslinker molecule that reacts with various functional groups, such as primary amines, thiols, phenols, and imidazoles, which are abundant in biopolymers. To use this strategy, a glutaraldehyde layer must be immobilized on the substrate, which can be achieved using a primer layer, such as amine-functionalized silane. Then, the glutaraldehyde-functionalized surface is incubated with the biopolymer, which spontaneously attaches to the crosslinker (Alavarse et al., 2022; Qureshi et al., 2020; Song et al., 2020). Carbodiimide is another crosslinker molecule that reacts specifically with carboxylic acid and primary amine groups, which are also common in biopolymers. To use this strategy, the substrate must be activated with a carbodiimide reagent, such as EDC, and a coupling agent, such as NHS, to form reactive esters. Then, the activated surface is incubated with the biopolymer, which reacts with the

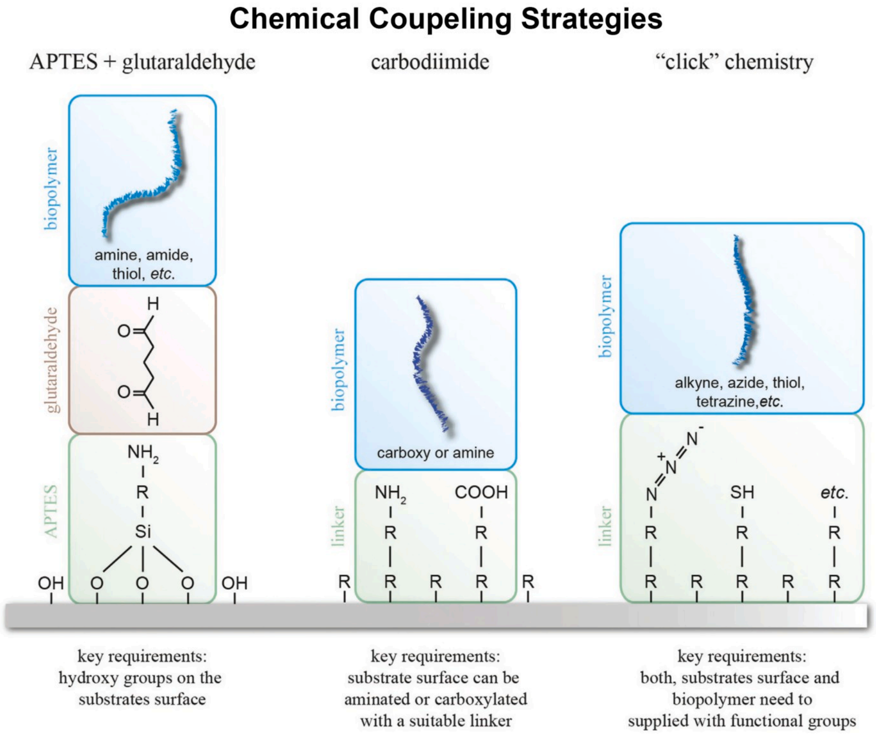

FIGURE 15.3 Common methods for the covalent immobilization of biopolymers on substrates. These involve intricate linking techniques utilizing (3-aminopropyl) triethoxysilane (APTES) and glutaraldehyde, carbodiimide coupling, or "click" chemistry.

These approaches utilize specialized linker molecules to target distinct functional groups on the biopolymers.

Reused from Song, J., Winkeljann, B., & Lieleg, O. (2020). Biopolymer-based coatings: Promising strategies to improve the biocompatibility and functionality of materials used in biomedical engineering. Advanced Materials Interfaces, 7*(17). https://doi.org/10.1002/admi.202000850. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.*

esters to form amide bonds (Alavarse et al., 2022; Qureshi et al., 2020; Song et al., 2020). Click chemistry is a term that refers to a group of reactions that are fast, selective, and orthogonal, meaning that they do not interfere with other reactions. One of the most popular click reactions is the copper-catalyzed azide-alkyne cycloaddition, which forms a triazole ring between an azide and an alkyne group. To use this strategy, either the biopolymer or the substrate must be modified with an azide or an alkyne group, and the other component needs to have the complementary group. Then, the azide- and alkyne-functionalized components are mixed in the presence of a copper catalyst, which triggers the click reaction (Alavarse et al., 2022; Qureshi et al., 2020; Song et al., 2020).

15.3.2.2 Polymer blending methods

Polymer blending is a pivotal aspect of polymer science and engineering, crucial for tailoring material properties for specific applications. Various methods are employed for polymer blending, each with its own set of advantages and limitations (Hoyos-Merlano et al., 2022; Khademian et al., 2020; Ndruru et al., 2021; Patel et al., 2022; Russo et al., 2021).

Mechanical blending is one of the most straightforward methods that involves physically mixing two or more polymers using devices, such as mixers, blenders, or extruders. While cost-effective, achieving a homogenous blend can be challenging, potentially leading to phase separation or uneven distribution of components. Common devices include twin-screw extruders and internal mixers, widely used in producing thermoplastic blends for improved mechanical, thermal, or barrier properties.

Solution blending involves the dissolution of polymers in a common solvent and subsequent solvent removal. Solution blending is effective for polymers with similar solubilities. Solvent selection is critical, considering compatibility and environmental impact. This method finds applications in producing biodegradable polymer blends for environmentally friendly purposes.

Melt blending, also known as melt mixing or melt processing, entails mixing polymers in a molten state, primarily used for thermoplastic polymers. Careful temperature control is necessary to ensure proper mixing without degradation. Melt blending is extensively used in the production of polymer alloys, combining properties for enhanced performance.

Reactive blending involves chemical reactions during blending. This method uses functional groups in polymer chains to form covalent bonds between different polymer components, resulting in a chemically modified blend. Reactive blending is commonly employed to enhance the toughness and impact resistance of polymers, making them suitable for demanding applications.

In-situ polymerization involves the simultaneous polymerization of monomers and blending of polymers, allowing for the synthesis of polymer blends with controlled molecular structures. Controlled radical polymerization, such as atom transfer radical polymerization, is a versatile method for in-situ polymerization, offering control over molecular weight and distribution. In-situ polymerization finds applications in producing conductive polymer blends for electronics and sensors.

15.3.3 Biopolymers as advanced coatings

Coatings and adhesives have been widely utilized in various industries to add or enhance specific material properties such as corrosion resistance, wear resistance, and conductivity (Singh & Singh, 2022). Among different materials with various origins and structures, polymers are by far the most used materials as coatings in diverse applications and segments. From simple coating by depositing thin films on materials to nanoparticle functionalized composite coatings or bioinspired tissue sealants, polymers, and biopolymers can serve proper and strong functionalities to their multitude of materials, such as metals, ceramics, polymers, and nanoparticles. In the biomedical field, biological coatings and adhesives have demonstrated their superiority in generating advanced biomaterials and instruments. They can join lightweight and different materials together to seal out moisture and air, speed production lines, and protect high-value and sensitive products.

Adhesives and sealants advance a whole host of technologies employed across a wide range of industries. Numerous reports discuss biopolymers as smart materials capable of mimicking biological environments while responding to numerous stimuli, such as temperature, light, magnetic field, electric field, and pH (Nathanael & Oh, 2020; Song et al., 2020). In biomedicine, these smart polymers are mainly used in drug delivery applications where the drugs can be loaded in these polymer or polymer coatings and can be delivered at the chosen location with the aid of a stimulus (Gehlen et al., 2019; Zhang & Chiao, 2015). Further, shape memory polymers are gaining interest and are promising candidates for bone tissue engineering applications (Azimi et al., 2020; Li et al., 2019; Sheikh et al., 2019). The integration and cooperation of polymeric materials with other polymers or inorganic materials led to the development of polymer-based composites with tunable properties, compensating for the problems associated with polymers in biomedical applications.

Like coatings, adhesives are indispensable in various types of applications. They are most outstanding in bio-applications where they can functionally be used as wound treatments and tissue regeneration (Bal-Ozturk et al., 2021; Ge & Chen, 2020; Zhu et al., 2018). Biopolymer-based bioadhesives with remarkable biocompatibility and biodegradability features could have garnered significant attention for real-time applications (Ma et al., 2016; Nam & Mooney, 2021; Reddy et al., 2021). As pioneers, polypeptides/proteins (with different types and sequences of amino acids as building units) and polysaccharides, such as chitosan and cellulose, with their unique molecular structures, offer excellent biocompatibility and are the two primary biopolymers employed in bioadhesive development (Bhagat & Becker, 2017; Rathi et al., 2019; Reggente et al., 2017; Sani et al., 2019). Recently, hydrogels as 3D polymer structures have been extensively studied as bioadhesives with promoted adhesion, biocompatibility, degradability, and antibacterial properties.

15.3.3.1 Commonly used biopolymers as coatings

15.3.3.1.1 Polyurethane

Polyurethane (PU) is a versatile synthetic polymer with notable advantages, including biodegradability, bioactivity, and flexibility, that is extensively utilized in diverse biomedical applications, such as coatings or directly as raw material in producing breast implants, vascular devices, and pacemaker leads. Going beyond its inherent properties, PU can effectively create composite materials by incorporating substances, such as nanomaterials. It was reported that encapsulation of silver lactate and sulfadiazine in PU results in the creation of bacterial resistance-PU coatings, which prevent the growth of microbial films (Roohpour et al., 2012). Playing with hard and soft segments in PU structures also leads to synthesizing different PUs with different properties. For example, Barrioni et al. reported the synthesis of catalyst-free PU films containing poly(caprolactone) triol and poly (ethylene glycol) as hard segments and hexamethylene diisocyanate and glycerol as soft segments (Barrioni et al., 2015). The final PU structure can be used as raw material in 3D printing different medical implants and equipment. Other than 3D printing of PU-based paste material, it has been shown that the creation of nanofibers by electrospinning the solution of different PU compositions can compete with conventional coatings. The electrospun fibers based on PU and graphene enhance PU's electroconductivity, bioactivity, and mechanical characteristics, thereby broadening its applicability in advanced biomedical settings (Bahrami et al., 2019). Furthermore, PU can undergo modifications with the addition of functional groups like polyethylene glycol (PEG) or zwitterion, effectively enhancing its antibacterial properties and corrosion resistance (Wang et al., 2019).

15.3.3.1.2 Polyvinylidene fluoride

As a fluoropolymer, polyvinylidene fluoride (PVDF) serves essential roles in tissue engineering, physiological signal detection, and developing antibacterial and antifouling materials (Chiu et al., 2013; Klinge et al., 2002). Boasting properties, such as inertness, water repellency, and piezoelectricity, PVDF can be skillfully applied to various substrates through techniques like spin coating and the Langmuir–Blodgett technique (Tien et al., 2014). Its potential is further maximized through synergistic combinations with nanomaterials like zinc oxide (ZnO), imparting augmented functionality to PVDF in biomedical applications (Chen et al., 2020; Shen et al., 2019; Shin et al., 2016; Tavakolmoghadam & Mohammadi, 2019).

15.3.3.1.3 Polypropylene

Acknowledged for its widespread use as a surgical mesh for tissue reinforcement, polypropylene (PP) stands out as a thermoplastic polymer with characteristics like low weight, inertness, and high mechanical strength (Saitaer et al., 2019). However, inherent challenges, such as high hydrophobicity, poor bioactivity, and susceptibility to inflammatory reactions necessitate surface modifications. The incorporation of materials like poly(styryl bisphosphonate) (poly(StBP))-6 addresses these challenges, resulting in improved biocompatibility and bioactivity (Steinmetz et al., 2020). Integrating poly(StBP) inside the coating enhances the optical features and durability of the coating material, which can be used in various applications involving bone tissue engineering.

15.3.3.1.4 Polydimethylsiloxane

A silicone-based polymer, polydimethylsiloxane (PDMS) finds extensive application in biomedical fields, including surgical implants, catheters, biosensors, drug delivery, and DNA sequencing (Steinmetz et al., 2020; Vargas et al., 2021). PDMS boasts many advantages, including bioactivity, adaptability, ease of fabrication, oxygen permeability, optical clarity, and low toxicity. Despite these advantages, PDMS has a limited capacity to interact with cells, which can be solved with surface modification (Sun, Wen, et al., 2020). PDMS can be modified similarly to PP with plasma treatment and the creation of hydroxyl groups. The generation of composite materials by combining PDMS materials, such as hyaluronic acid (HA), to enhance hemocompatibility and antifouling properties can expand its range of biomedical applications (Saboktakin et al., 2011; Xue et al., 2017).

15.3.3.1.5 Polymethyl methacrylate

Polymethyl methacrylate (PMMA) is a lightweight and transparent polymer used in various medical applications, such as drug delivery, bone cement, microsensors, denture bases, and implants (Reggente et al., 2018). Renowned for its inertness, high mechanical properties, moderate degradation rate, and low toxicity, PMMA can be effectively coated onto metal surfaces, like titanium and zinc, enhancing the adhesion and formability (Mousa et al., 2018). Moreover, the incorporation of ZnO nanoparticles further enhances PMMA's antibacterial activity and corrosion resistance, offering advanced solutions in medical material science (Reggente et al., 2018).

In addition to the synthetic polymers mentioned above, natural polymers like collagen and chitosan play pivotal roles in biomedical coatings. These natural polymers offer intrinsic benefits, such as bioactivity, biodegradability, and low toxicity. Expanding their versatility, these polymers can also be employed for coating nanoparticles, amplifying their applications in drug delivery and other biological contexts (Fig. 15.4) (Avcu et al., 2019; Farrokhi-Rad et al., 2017; Frank et al., 2020; Wei & Haag, 2015). The amalgamation of synthetic and natural polymer coatings not only underscores their diverse functionalities but also signifies their paramount contribution to the ever-evolving landscape of biomedical materials. Among the biopolymers, structures including poly(lactide-*co*-glycolic), polycaprolactone, poly(lactic acid), and polyethylene are regularly employed in biological and medical applications. Poly(lactic acid) is extensively applied in various biomedical fields, such as tissue engineering, drug delivery, and the fabrication of 3D printing scaffolds (Singh & Singh, 2022). The incorporation of ZnO nanoparticles into poly(lactic acid) proves beneficial, contributing to the control of degradation and augmentation of antibacterial activity. It allows the manipulation of surface topography and the rate of magnesium breakdown. The use of poly(L-lactide) (PLLA) solution on PDMS results in the creation of PLLA microchambers, facilitating drug loading and sealing, thereby enhancing drug delivery outcomes (Clavijo et al., 2016).

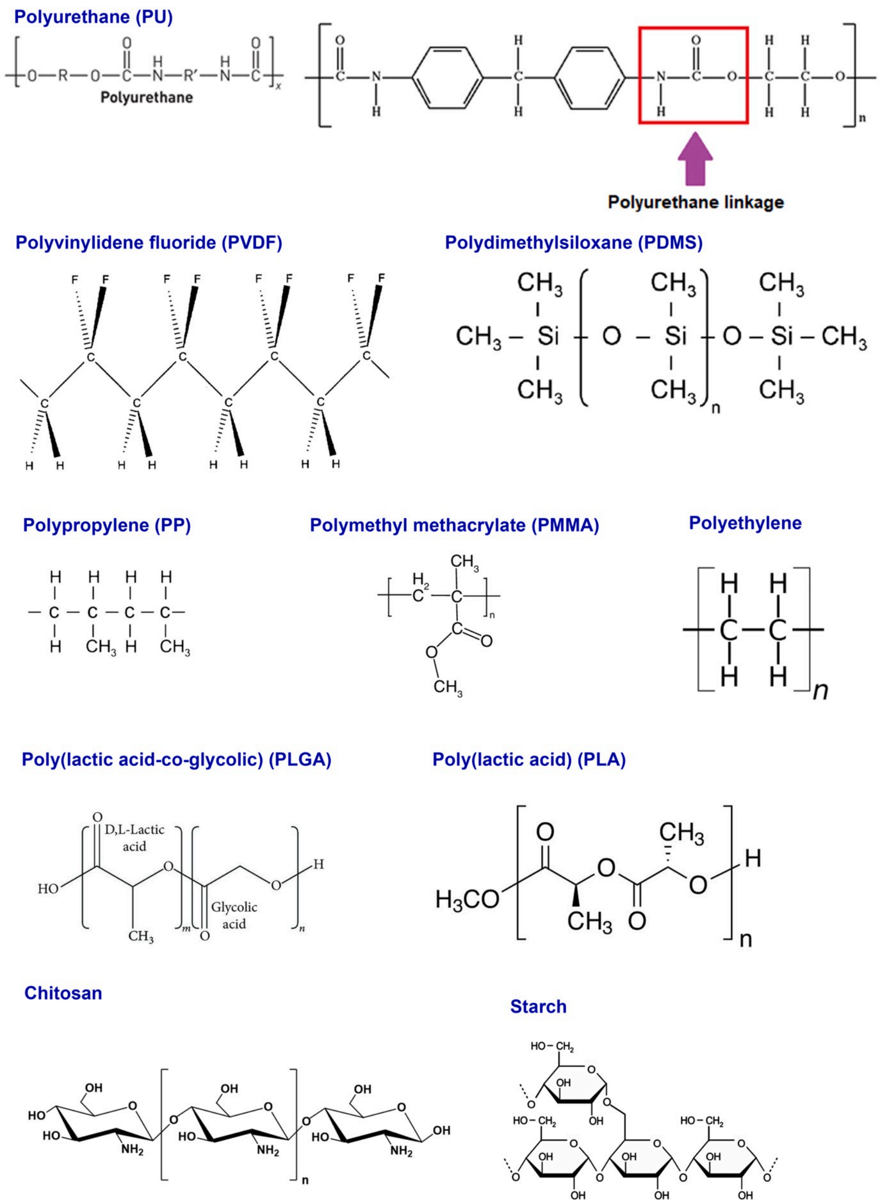

FIGURE 15.4 Example of the chemical compositions of frequently employed biopolymer in coatings.

The polymer structures presented include polyurethane (PU), polyvinylidene fluoride (PVDF), polypropylene (PP), polydimethylsiloxane (PDMS), polymethyl methacrylate (PMMA), poly(lactide-*co*-glycolic) (PLGA), and poly(lactic acid) (PLA).

15.3.3.2 Biomedical application of biopolymer-based coatings

Biopolymer-based coatings can be used in various biomedical applications, which can be categorized into four main types: biotribology and antibiofouling, promoting cellular adhesion, drug delivery, and biosensing. Biotribology and antibiofouling coatings aim to reduce the friction, wear, and corrosion of the materials and prevent the unwanted adhesion of proteins, pathogens, or cells. These coatings can enhance the lubricity, wettability, and fouling resistance of the materials and thus extend their lifetime and functionality (Jahn & Klein, 2015; Seo et al., 2018). Some of the biopolymers that have been used to create lubricious and antibiofouling coatings include pectin, lubricin, chitosan, mucin, and poly-L-lysine (Benz et al., 2004; Gu et al., 2009; Ishihara, 2015; Niemczyk et al., 2015). These biopolymers can be either physically or chemically attached to the surface, depending on the strength and stability of the coating required. For example, mucin and lubricin can be covalently coupled to the surface via carbodiimide chemistry, while chitosan and poly-L-lysine can be physically adsorbed to the surface via electrostatic interactions.

Another goal for biopolymer-based coatings is to enhance the cell-adhesive properties of the material surface, which is essential for tissue engineering applications. Here, the biopolymers can act as scaffolds or matrices that support cell attachment, migration, proliferation, and differentiation. The biopolymers can also provide biochemical cues or signals that modulate cellular behavior and function (Huang et al., 2016). Some of the biopolymers that have been used to promote cellular adhesion include poly(dopamine), HA, and chitosan. Depending on the desired coating thickness and properties, these biopolymers can also be physically or chemically deposited on the surface. For example, poly(dopamine) can be spontaneously formed on the surface by oxidative polymerization of dopamine (DA), while HA and chitosan can be covalently grafted to the surface via click chemistry or carbodiimide chemistry (Li, Chen, et al., 2012; Li, Jiang, et al., 2012; Silva et al., 2017; Veisi et al., 2019; Zheng et al., 2019).

Biopolymer-based coatings enable the controlled release of drugs or other bioactive molecules from the material surface. This can be useful for wound healing, infection prevention, or cancer therapy applications. Biopolymers can act as carriers or reservoirs that encapsulate or immobilize the drugs and release them in a controlled manner in response to external stimuli or environmental changes (Alibolandi et al., 2017; Moscariello et al., 2019; Yucel Falco et al., 2019). Biopolymers, such as alginate, heparin, and gelatin have been widely used for drug delivery coatings. These biopolymers can form hydrogels or films on the surface, which can be loaded with drugs or other molecules and the drugs release can be modulated by factors, such as pH, temperature, or enzymes. Moreover, biopolymer-based coatings can be utilized to detect and quantify biomolecules or biochemical reactions on the surface. This can be useful for diagnostics, monitoring, or imaging applications. At the same time, biopolymers can act as receptors or catalysts that bind or react with the target molecules and generate a measurable signal, such as optical, electrical, or magnetic (Ronkainen et al., 2010; Sin et al., 2014). Some of the biopolymers that have been used for biosensing coatings include DNA, RNA, and enzymes. These biopolymers can be immobilized on the surface via physical or chemical methods and can recognize or transform the target molecules with high specificity and sensitivity. For example, DNA can form hybridization complexes with complementary DNA or RNA on the surface and generate a fluorescent or colorimetric signal. RNA can form aptamer complexes with proteins or small molecules on the surface and generate an electrochemical or optical signal (Geng et al., 2011; Heckman et al., 2011). Enzymes can catalyze reactions with substrates or inhibitors on the surface and generate a colorimetric or electrochemical signal (Abdullah et al., 2006).

15.3.3.3 Application in other industries

Biopolymer-based coatings have diverse applications in the food industry, extending beyond the biomedical field. They find utility in smart packaging, food preservation, and quality control. In smart packaging, biopolymer coatings can modify surfaces to incorporate biosensors, detecting contaminants or spoilage indicators, ensuring enhanced food safety (Fig. 15.5) (Popescu & Ungureanu, 2023). Additionally, these coatings contribute to food preservation by extending shelf life and inhibiting microbial growth. Active packaging materials with biopolymer coatings containing nanoparticles like silver, copper, or ZnO exhibit strong antimicrobial activity against various spoilage-causing microorganisms. Furthermore, biopolymer coatings, enriched with essential oils, antioxidants, or preservatives, prevent oxidation and degradation of food components. Another vital application is in food quality control, where biopolymer coatings enable the creation of biosensors for analyzing the chemical composition of food products and detecting harmful substances. This approach provides a rapid, accurate, and reliable method for food analysis, bypassing the complexities of traditional laboratory techniques (Popescu & Ungureanu, 2023). Beyond food applications, integrating antimicrobial agents in edible films, particularly those based on biodegradable polymer films, is important in extending product shelf life while adhering to sustainable and environmentally friendly packaging practices (Anvar et al., 2021).

The automobile and aviation industries also benefit from using biopolymers in engineering applications. Biopolymers, such as polyaniline, polypyrrole, and polythiophene are used to coat metal surfaces for corrosion protection. These coatings can form a conductive and protective layer on the metal surface and act as a barrier to inhibit the diffusion of corrosive agents (Rahman et al., 2023). For example, a study described the development of a nanocomposite biopolymer coating using $ZnO–TiO_2–PLE$ as a protective material for carbon steel against corrosion in both acidic and marine environments. They showed that the biopolymer coating outperforms uncoated steel and steel coated with epoxy resin (DGEBA) in terms of corrosion resistance, hydrophobicity, thermal stability, and mechanical strength (Murugesan et al., 2023). The biopolymer-based coating materials represent a promising and environmentally friendly solution for safeguarding carbon steel against corrosion in aggressive environments.

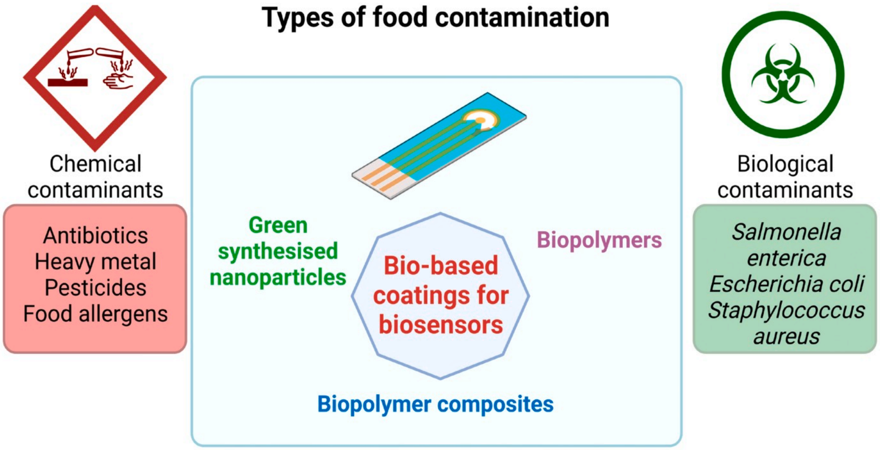

FIGURE 15.5 Application of biocoatings in the development of food quality biosensors.

Reused from Popescu, M., & Ungureanu, C. (2023). Biosensors in food and healthcare industries: Bio-coatings based on biogenic nanoparticles and biopolymers. Coatings, 13*(3).* *https://doi.org/10.3390/coatings13030486. © 2023.*

Biopolymers can also be used as sustainable and functional wood coatings. A study showed the feasibility of substituting fossil-based polyol with lignin to create flame-retardant PU. The aromatic structure of lignin contributes to forming a dense carbon layer in the PU matrix, acting as a barrier against combustible gases and blocking oxygen, thus reducing the flame rate (Wang et al., 2020). However, formulations with a high lignin percentage may lead to cracking and increased permeability compared to petrochemical counterparts. Adding low-surface-tension film-forming agents presents a viable solution to mitigate cracking (Bergamasco et al., 2022). Despite lignin's potential to enhance biobased content in PU coatings, challenges, such as low solubility, reactivity, high glass transition temperature (Tg), polydispersity, photodegrading propensity, and dark color persist. Further research is essential to address these issues and facilitate the widespread integration of lignin in PUs (Landry et al., 2023).

15.4 Applications of functionalized biopolymers in adhesives

Biopolymer-based adhesives are materials that can bond to biological tissues or surfaces using natural polymers as the main components with benefits, such as good biocompatibility, biodegradability, and low immunogenicity (Bruschi & Freitas, 2005; Lee et al., 2000). Polypeptides, proteins, and polysaccharides derived from animals, plants, or microorganisms are well-known examples of biopolymers that have been

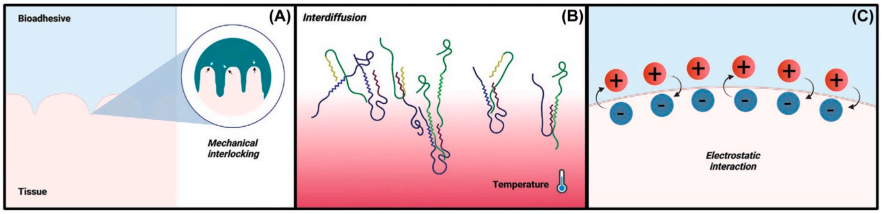

FIGURE 15.6 Exploring primary adhesion mechanisms.

(A) Mechanical interlocking, (B) interdiffusion, and (C) electrostatic interaction in tissue-bioadhesive interfaces.

Reused with permission from Li, J., Yu, X., Martinez, E. E., Zhu, J., Wang, T., Shi, S., Shin, S. R., Hassan, S., & Guo, C. (2022). Emerging biopolymer-based bioadhesives. Macromolecular Bioscience, 22*(2). https://doi.org/10.1002/mabi.202100340. © 2021 Wiley-VCH GmbH.*

employed as bioadhesives (Cazacu et al., 2012; Nam & Mooney, 2021). These biopolymers can be modified or cross-linked to achieve desired mechanical properties, adhesion strength, and degradation rate. Some examples of biopolymer-based adhesives are fibrin, albumin, gelatin, silk, chitosan, alginate, and HA (Gowda et al., 2020; Jia et al., 2021; Jiang et al., 2021; Jin et al., 2021). In addition to these biopolymers, other natural materials, such as glycoproteins, xanthan gum, and tannic acid have also been explored for developing bioadhesives. Biopolymer-based adhesives can also incorporate various functions, such as hemostasis, antimicrobial activity, tissue repair, and biomineralization. Furthermore, biopolymer-based adhesives can be designed to respond to different stimuli, such as pH, temperature, light, and enzymes, to achieve controlled adhesion and release (Li et al., 2022).

The two main methods involving physical or mechanical approaches and chemical technologies have been used to achieve strong bonding between bioadhesives and their surfaces (Fig. 15.6). Mechanical adhesion relies on the physical interactions between the adhesive and the tissue surface, such as mechanical interlocking, chain entanglement, and electrostatic attraction. On the other side, chemical adhesion involves the formation of intermolecular bonds, such as covalent, ionic, and hydrogen bonds, between the adhesive and the desired molecules. Both mechanisms can be influenced by various factors, such as the surface roughness, the molecular weight, the charge density, and the environmental conditions of the adhesive and the desired molecules or surfaces. Combining the two methods is applicable for some bioadhesives to enhance the adhesion performance. For example, some bioadhesives inspired by natural organisms, such as mussels and spiders, can use catechol groups or glycoproteins to regulate the interfacial water and form reversible or irreversible bonds with the tissue surface.

15.4.1 Physical adhesion approaches

Physical adhesion involves the formation of noncovalent bonds or interactions between the bioadhesive and the desired surface. Depending on the nature of the adhesive and the tissue, different types of adhesion mechanisms can be involved in bioadhesives. Some of the main mechanisms include mechanical interlocking, interdiffusion, and electrostatic interaction.

Mechanical interlocking occurs when the adhesive fills the micro-irregularities and pores of the desired surface and forms a mechanical lock. The roughness and porosity of the surface affect the degree of interlocking. This mechanism is often combined with others, such as physical or chemical cross-linking, to enhance the adhesion strength and stability. For example, poly(styrene) and poly(acrylic acid)-based microneedles with the ability to penetrate the surface, followed by swelling in the presence of liquid, can lead to local deformation and consequently interlocking with the surface.

Interdiffusion or chain entanglement happens when the polymer chains of the bioadhesive and the tissue intermingle or entangle with each other, forming a network that bridges the interface. The degree of entanglement depends on the molecular weight, cross-linking density, and diffusion coefficient of the adhesive. One good example of such a mechanism is related to mucoadhesion, where the adhesive and the mucin glycoproteins interpenetrate and entangle. Electrostatic bonding takes place when there is a difference in charge between the adhesive and the tissue, resulting in an attraction between them by coulombic forces and the creation of an electric double-layered interface. The strength of the adhesive (the resistance to separation) depends on the attractive forces present in the double layer. Other than the explained physical methods, surface wetting is another approach that can make a proper impact in the case of correct use. Surface wetting is the ability of the adhesive to spread and make contact with the desired surface. Wetting is essential for the formation of intermolecular bonds and the displacement of interfacial water. The wetting behavior of the adhesive depends on its surface tension and the critical surface tension of the surface. A good wetting is achieved when the contact angle between the adhesive and the surface is low (Modaresifar et al., 2016; Pandey et al., 2020). It is worth mentioning that physical adhesion is usually reversible and depends on factors, such as surface roughness, contact time, temperature, and pressure.

15.4.2 Chemical adhesion approaches

Unlike physical methods, chemical adhesion depends on the intermolecular forces that strongly bond the bioadhesive and the surface. The intermolecular forces can be classified into primary and secondary forces, and combining these two forces is also possible. The primary forces include ionic, covalent, and metallic bonds, while the secondary forces involve dipole–dipole interactions, London dispersion forces, and van der Waals forces (Fig. 15.7). Nature always illustrates the best examples in this matter. For example, some natural organisms, such as geckos, spiders, mussels, and slugs, could develop remarkable adhesion strategies based on chemical adhesion. These natural examples inspire researchers to design bioadhesives with enhanced wet-adhesion properties.

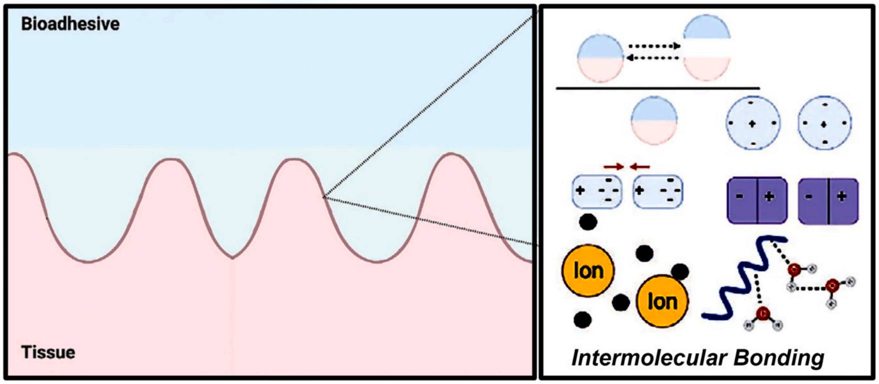

FIGURE 15.7 Intermolecular bonding

It includes reversible and irreversible bonds, hydrogen bonding, coordination bonding, and van der Waals forces.

Reused with permission from Li, J., Yu, X., Martinez, E. E., Zhu, J., Wang, T., Shi, S., Shin, S. R., Hassan, S., & Guo, C. (2022). Emerging biopolymer-based bioadhesives. Macromolecular Bioscience, *22(2). https://doi.org/10.1002/mabi.202100340. © 2021 Wiley-VCH GmbH.*

One of the most widely studied bioinspired strategies is based on the mussel adhesive proteins (MAPs), which contain high amounts of L-3,4-dihydroxyphenylalanine (DOPA) residues. DOPA can form strong and reversible bonds with various substrates through catechol chemistry, which involves hydrogen bonding, electrostatic interactions, and metal chelation (Fig. 15.8A and B) (Cui et al., 2020; Luo et al., 2020). To mimic the MAPs, researchers have modified various biopolymers with DOPA or DA groups, such as cellulose, chitosan, HA, gelatin, and silk fibroin (SF). These DOPA-modified biopolymers have shown improved wet-adhesion performance on various tissues, such as skin, liver, heart, and kidney.

Besides DOPA and DA, other molecules that can form similar bonds with substrates have also been explored for bioadhesive design, such as boronic acid (Hong et al., 2020), caffeic acid (Thirupathi et al., 2013), and tannic acid (Ke et al., 2020). Boronic acid can form reversible covalent bonds with *cis*-diol groups on the tissue surface through boronate ester formation (Fig. 15.9). Caffeic acid and tannic acid are phenolic compounds that can form hydrogen bonds and metal chelation with substrates. These molecules have been used to modify biopolymers, such as gelatine and SF to enhance their adhesion properties (Fig. 15.10).

Another bioinspired strategy is based on the adhesive secretions of algae, which can adhere to wet surfaces by producing polyphenol aggregates that can displace water molecules and form hydrogen bonds with substrates. Researchers have developed alginate-based bioadhesives by incorporating polyphenols, such as DA and ferric ions to mimic the algae adhesion mechanism (Fig. 15.11) (Cholewinski et al., 2019).

PEG is a synthetic polymer approved by the FDA for biomedical applications due to its biocompatibility and low immunogenicity. PEGylation, which refers to

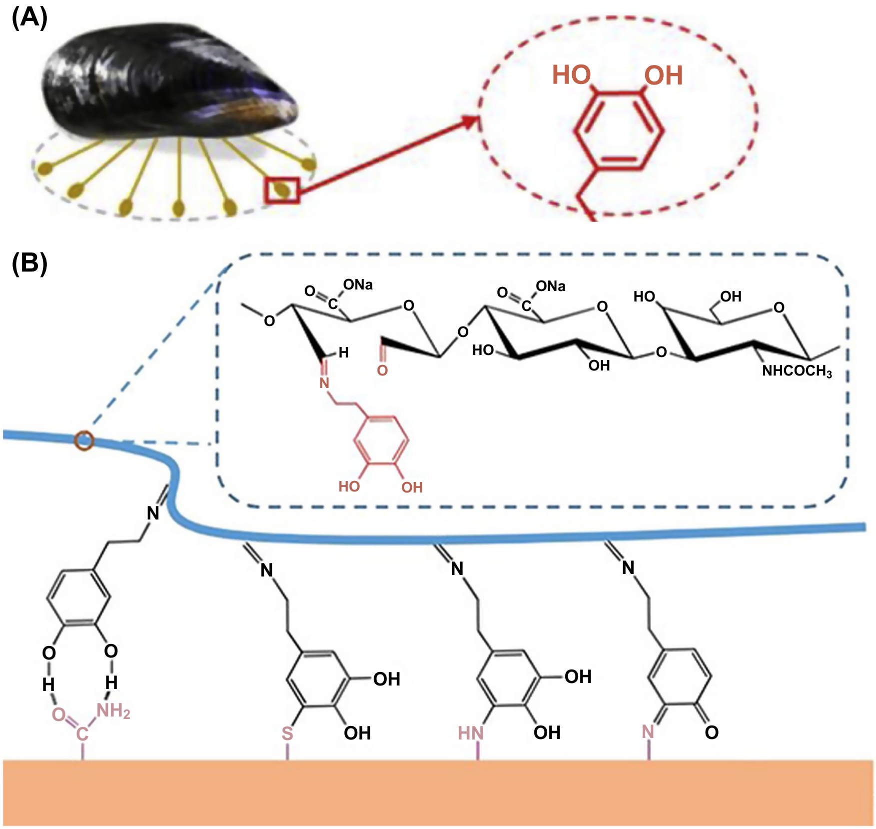

FIGURE 15.8 Varied strategies for enhancing wet-adhesion capability.

(A) Distinct structure in mussel adhesive proteins (Zhong et al., 2019). (B) Dopamine-modified hyaluronic acid (HA) bioadhesives (Zhou et al., 2020).

Reproduced with permission from (A) Zhong, Y., Wang, J., Yuan, Z., Wang, Y., Xi, Z., Li, L., Liu, Z., & Guo, X. (2019). A mussel-inspired carboxymethyl cellulose hydrogel with enhanced adhesiveness through enzymatic crosslinking. Colloids and Surfaces B: Biointerfaces, 179, *462–469. http://doi.org/10.1016/j.colsurfb.2019.03.044. Available from http://www.elsevier.com/locate/colsurfb. © 2019; (B) Zhou, D., Li, S., Pei, M., Yang, H., Gu, S., Tao, Y., Ye, D., Zhou, Y., Xu, W., & Xiao, P. (2020). Dopamine-modified hyaluronic acid hydrogel adhesives with fast-forming and high tissue adhesion.* ACS Applied Materials & Interfaces, 12*(16), 18225–18234. https://doi.org/10.1021/acsami.9b22120. © 2020 American Chemical Society.*

the conjugation of PEG chains to biomolecules, can improve the solubility, stability, and functionality of bioadhesives. For example, PEG-modified SF bioadhesives have shown enhanced wet-adhesion strength and electrical conductivity, making them suitable for biosensing applications (Fig. 15.12) (Liu et al., 2020).

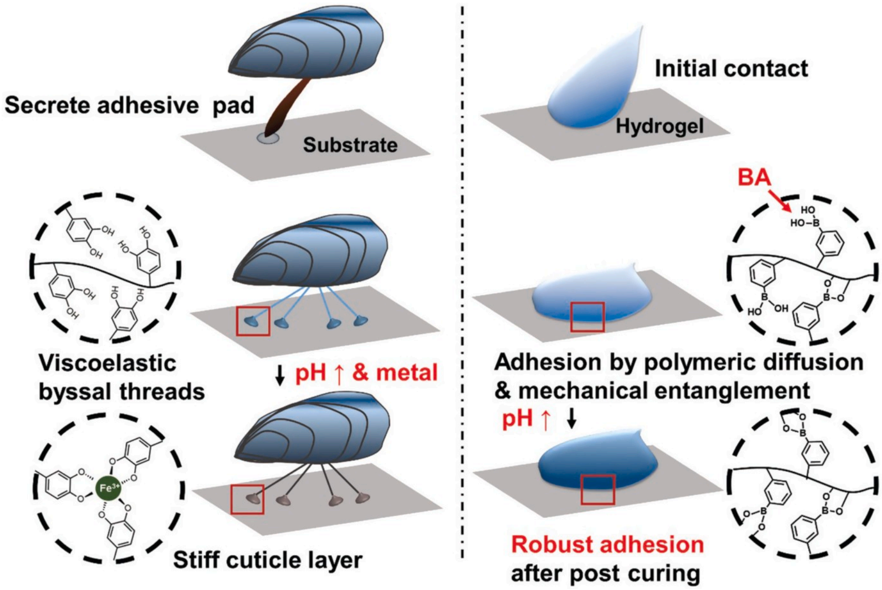

FIGURE 15.9 Varied strategies for enhancing wet-adhesion capability.

Catechol-containing mussel threads' adhesion mechanism and bioinspired boronic acid-containing bioadhesive.

Reused with permission from Hong, S. H., Shin, M., Park, E., Ryu, J. H., Burdick, J. A., & Lee, H. (2020). Alginate-boronic acid: pH-triggered bioinspired glue for hydrogel assembly. Advanced Functional Materials, 30*(26). https://doi.org/10.1002/adfm.201908497. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.*

15.4.3 Commonly used biopolymers as adhesives

15.4.3.1 Polyesters

Polyesters (PS) are polymers containing ester linkages in their backbone and are prone to hydrolytic cleavage (Fig. 15.13). These types of synthetic polymers can be synthesized from various monomers, such as lactic acid, glycolic acid, caprolactone, and dioxanone, to obtain different copolymers with varying properties, such as molecular weight, crystallinity, hydrophilicity, and degradation rate. PS are widely used for controlled drug release, proteins, and genes and scaffolds for tissue regeneration.

15.4.3.2 Polyanhydrides

Polyanhydrides are polymers that contain anhydride linkages in their backbone, which are also susceptible to hydrolysis. Polyanhydrides can be synthesized from various diacids, such as sebacic acid, adipic acid, and terephthalic acid, and copolymerized with other monomers, such as esters and amides, to modulate their

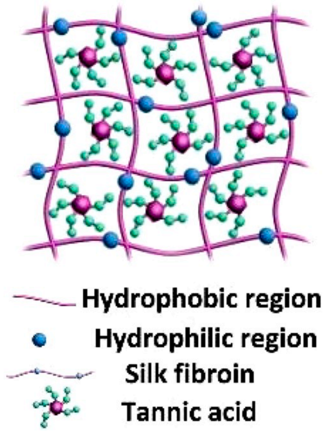

FIGURE 15.10 Varied strategies for enhancing wet-adhesion capability.

Silk fibroin and tannic acid bioadhesive structure (Gao et al., 2020).

Reused with permission from Gao, X., Dai, Q., Yao, L., Dong, H., Li, Q., & Cao, X. (2020). A medical adhesive used in a wet environment by blending tannic acid and silk fibroin. Biomaterials Science, 8*(9), 2694–2701. https://doi.org/10.1039/d0bm00322k. © 2020 Royal Society of Chemistry.*

properties, such as erosion rate, surface erosion, and drug release kinetics. Polyanhydrides are mainly used for localized delivery of drugs, especially anticancer agents, to tumors and other pathological sites.

15.4.3.3 Polyamides

Polyamides (PIm) contain amide linkages in their backbone, which are more resistant to hydrolysis than esters and anhydrides. PIm can be synthesized from various amino acids, such as iminocarbonates and polyamino acids, to obtain polymers with different chain configurations, such as linear, branched, or cyclic. PIm, such as sutures, stents, and implants, are used for drug delivery applications that require high mechanical strength and stability.

15.4.3.4 Phosphorus-based polymers

Phosphorus-based polymers contain phosphorus atoms in their backbone, which can impart unique properties, such as flame retardancy, biocompatibility, and immunomodulation. Phosphorus-based polymers can be synthesized from various monomers, such as phosphates, phosphonates, and phosphazenes, to obtain polymers with different functional groups, such as hydroxyl, carboxyl, and amino groups. Phosphorus-based polymers are used for the delivery of drugs, proteins, and vaccines, as well as for the modulation of immune responses and inflammation.

Other than the above-mentioned synthetic polymers, some do not belong to the above categories but have been used for bioadhesive drug delivery systems, such as

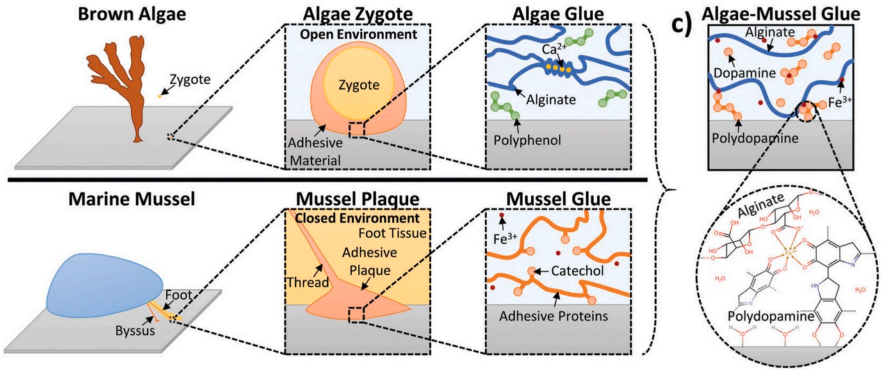

FIGURE 15.11 Varied strategies for enhancing wet-adhesion capability.

Algae–mussel hydrogel bioadhesive with algae and mussel adhesion mechanisms.

Reused with permission from Cholewinski, A., Yang, F., & Zhao, B. (2019). Algae-mussel-inspired hydrogel composite glue for underwater bonding. Materials Horizons, *6(2), 285–293. https://doi.org/10.1039/c8mh01421c, http://pubs.rsc.org/en/journals/journal/mh. © 2019 Royal Society of Chemistry.*

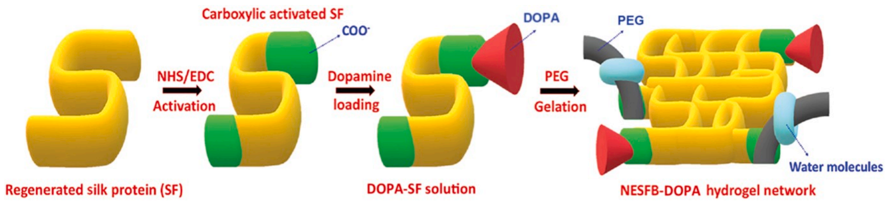

FIGURE 15.12 Varied strategies for enhancing wet-adhesion capability.

PEGylation process of DOPA-SF and bioadhesive networks.

Reused with permission from Liu, H., Yuan, M., Sonamuthu, J., Yan, S., Huang, W., Cai, Y., & Yao, J. (2020). A dopamine-functionalized aqueous-based silk protein hydrogel bioadhesive for biomedical wound closure. New Journal of Chemistry, 44*(3), 884–891. https://doi.org/10.1039/c9nj04545g, http://pubs.rsc.org/en/journals/journal/nj. © 2020 Royal Society of Chemistry.*

polyacrylates, PUs, polyorthoesters, polyacetals, and PEG. Depending on their chemical structure and composition, these polymers can have various properties, such as hydrophilicity, hydrophobicity, biodegradability, biostability, and mucoadhesion. These polymers can be used for various routes of administration, such as oral, nasal, ocular, buccal, vaginal, and rectal.

15.4.4 Biomedical application of biopolymer-based adhesives

Bioadhesives are superior biomaterials used in different biomedical applications. These materials can be useful for sealing wounds, stopping bleeding, and providing other

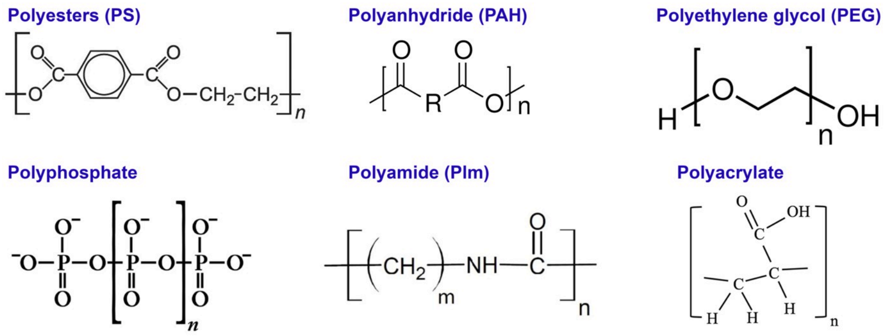

FIGURE 15.13 Examples of polymer structures used in biopolymer-based adhesives

The depicted polymers are polyester (PS), polyanhydride (PAH), polyamide (PIm), phosphorus-based polymers, polyacrylate, and polyethylene glycol (PEG).

biological functions that can enhance healing and prevent infections. Moreover, the chemistry of biopolymers allows the design and generation of different structures that can interact with various biological tissues and organs, such as skin, bone, cartilage, cornea, and blood vessels, for different biomedical applications. Biopolymer-based bioadhesives can act as hemostatic agents by forming a physical barrier, activating the coagulation cascade, or enhancing platelet aggregation. Many bioadhesives are based on fibrin, gelatin, or albumin, which can interact with blood components and form a fibrin clot to achieve hemostasis. For example, Evicel is a fibrin-based bioadhesive that uses thrombin to convert fibrinogen into fibrin, which can adhere to the wound site and stop bleeding (Li et al., 2022). BioGlue is an albumin-based bioadhesive that uses glutaraldehyde to crosslink albumin and tissue proteins, forming a strong bond and preventing blood leakage (Tarafder et al., 2020). ProGel is another albumin-based bioadhesive that uses a synthetic crosslinker to form amide bonds between albumin and tissue, resulting in a flexible and durable sealant. Gelatin-based bioadhesives, such as FloSeal and Surgiflo, consist of gelatin granules and thrombin, which can swell and absorb blood, activate the coagulation cascade, and form a fibrin matrix. Other bioadhesives, such as chitosan, alginate, and carboxymethyl cellulose (CMC), can also exhibit hemostatic properties by interacting with blood components, such as red blood cells, platelets, and plasma proteins. For instance, HemCon is a chitosan-based bioadhesive that can form electrostatic interactions with negatively charged blood cells and proteins, leading to rapid clot formation and bacterial inhibition (Li et al., 2022). Silvercel is an alginate-based bioadhesive that contains silver-coated nylon fibers, which can release silver ions and exert antimicrobial and antiinflammatory effects, as well as absorb wound exudate and promote hemostasis (Gray, 2009; Morozkina et al., 2022). CMC is a polysaccharide derivative that can swell and form a gel-like structure upon contact with blood, creating a mechanical barrier and reducing blood flow. The hemostatic properties of CMC may also be attributed to its acidic pH and negative charge, which can stimulate platelet activation and aggregation. In addition to hemostasis and wound closure, bioadhesives can also support tissue repair and regeneration by providing a scaffold, delivering growth factors, or stimulating cellular responses. For example, a polydopamine-chondroitin sulfate-polyacrylamide hydrogel bioadhesive was developed for cartilage regeneration (Han et al., 2018). The bioadhesive showed good adhesion to cartilage tissue and enhanced chondrogenic differentiation of stem cells under growth factor-free conditions.

Another example is GelCORE, a gelatin-based bioadhesive designed for corneal regeneration (Sani et al., 2019). The bioadhesive was transparent, biocompatible, tunable, and could facilitate corneal epithelial and stromal healing in a rabbit model. Bioadhesives with biomineralization properties can be used for bone tissue repair by inducing the formation of hydroxyapatite crystals, which are the main inorganic component of bone. For instance, a DA-modified gelatin (Gel-DA) bioadhesive was prepared to enhance the biomineralization of gelatin (Zhang et al., 2021). The bioadhesive could form a Gel-DA@AgNPs precursor solution, which could be further cross-linked with guar gum and boric acid to form a Gel-DA/GG@Ag hydrogel with antibacterial and osteogenic properties. Another example is a bioadhesive composed of bael fruit gum, chitosan, and nano-hydroxyapatite, which showed good adhesion, biocompatibility, and biomineralization for bone tissue engineering (Mirza et al., 2018).

Bioadhesives can also be used as substrates or matrices for bioelectronics, especially epidermal bioelectronics, which are flexible and conformable devices that can interface with the skin and monitor or modulate physiological signals. Bioadhesives can provide mechanical support, electrical conductivity, biocompatibility, and biodegradability for bioelectronic devices. For example, SF has been used to develop adhesive bioelectronics, such as a silk-based hybrid bioelectronic device that combined SF and poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) for biosensing applications, such as strain sensing (Sharma et al., 2019). Another example is a calcium-modified SF bioadhesive that showed enhanced viscoelasticity and adhesion to the skin and could be used for epidermal electronics, such as electrocardiogram monitoring (Jahn & Klein, 2015).

15.4.5 Biopolymers as innovative adhesives

Biopolymers have emerged as promising alternatives to traditional adhesives, offering sustainable and eco-friendly solutions across various industries. Using natural biopolymers in adhesive formulations represents a notable innovation in the field. One significant trend involves harnessing natural biopolymers for adhesive applications. Starch, derived from crops like corn or potatoes, has gained attention for its abundance, low cost, and renewability. The biodegradable nature of starch-based adhesives makes them environmentally friendly. Additionally, gelatin, a protein derived from animal collagen or plant-based sources, has been explored for its adhesive properties. These natural biopolymers present opportunities to develop adhesives with reduced environmental impact. Recent advancements in biopolymer-based adhesives focus on enhancing biocompatibility, especially in medical and healthcare applications. Biocompatible adhesives are designed to interact harmoniously with biological systems, minimizing potential adverse effects. For instance, chitosan, a derivative of chitin found in crustacean shells, has been employed in medical adhesives due to its biocompatibility and antimicrobial properties. Such innovations pave the way for safer and more effective adhesive solutions in sensitive applications. The quest for eco-friendly adhesive solutions has led to the development of adhesives derived from renewable resources. Lignin, a byproduct of the paper and pulp industry, has garnered attention for its potential as a biobased adhesive. Lignin-based adhesives not only utilize a waste product but also reduce reliance on petrochemical-derived adhesives, contributing to a more sustainable adhesive industry. This trend aligns with the growing demand for environmentally conscious products and processes. Various materials have gained prominence for their eco-friendly attributes. Starch-based adhesives, known for their ease of processability and compatibility with other polymers, are employed in diverse applications, such as packaging, textiles, and labeling. Gelatine-based adhesives find applications in biomedical fields, including drug delivery systems and tissue engineering, owing to their biocompatibility and film-forming properties. Lignin, a complex aromatic polymer, is being explored for its adhesive potential in plywood, particleboard, and other wood products, contributing to the circular economy by repurposing a significant industrial byproduct (Antosik et al., 2019; Chen et al., 2021; Cui et al., 2020; Ruwoldt et al., 2023; Sharma et al., 2021; Vrabic-Brodnjak, 2023).

15.5 Challenges and future directions

Biopolymers-based coatings and adhesives are promising strategies to improve the biocompatibility and functionality of materials used in biomedical engineering. Several examples of biopolymer coatings have been demonstrated in the areas of implants/biomedical devices, tissue engineering, drug delivery, and biosensing, showing the potential and versatility of this approach. However, biopolymer-based coatings still face some challenges and limitations that need to be addressed in the future. Some of these challenges include the cost and availability of biopolymer sources, the stability and durability of biopolymer coatings in physiological environments, the control and optimization of the coating properties and performance, the standardization and regulation of the coating methods and applications, and the evaluation and assessment of the biocompatibility and biodegradability of biopolymer coatings. To overcome these challenges, more research and development efforts are needed to explore new sources and types of biopolymers, to design novel and efficient coating methods and techniques, to characterize and tailor the coating properties and functions, to establish and follow the quality and safety standards and guidelines, and to test and validate the coating outcomes and impacts. Similar to biopolymer-based coatings, bioadhesives with biopolymer origins have been examined and used for various biomedical purposes, including tissue adhesion, hemostasis, wound healing, and bioelectronics. Despite all their outstanding features, there are still some challenges and limitations that need to be addressed, such as improving the wet-adhesion strength and durability of bioadhesives to different types of tissues and organs under physiological conditions (Lakshmipriya et al., 2023; Perera et al., 2023; Sing et al., 2022; Sujatha & O'Kelly, 2023; Thomas & Melethil, 2023).

To overcome these challenges and realize the full potential of biopolymer-based coatings and bioadhesives, more interdisciplinary and collaborative efforts are needed among materials scientists, biologists, engineers, clinicians, and regulatory agencies. Some of the goals that need to be addressed are enhancing the mechanical properties and compatibility of bioadhesives to the dynamic and complex tissue environments, reducing the potential risks of infection, inflammation, and immunogenicity caused by bioadhesives or their degradation products, developing bioadhesives with smart and responsive features, such as self-healing, stimuli-responsiveness, and bioactivity, to adapt to the changing tissue needs and functions, optimizing the fabrication and processing methods of bioadhesives to ensure their stability, scalability, and cost-effectiveness, evaluating the long-term safety and efficacy of bioadhesives in preclinical and clinical trials. As biopolymer applications continue to evolve, future directions hold promising avenues for innovative developments. The intersection of biopolymers with various technologies opens up new possibilities, and several emerging trends are poised to shape the landscape.

15.5.1 Smart coating and hybrid materials innovations

The future of biopolymer applications envisions the integration of smart coating and adhesive technologies. These smart materials can respond to environmental stimuli and offer functionalities, such as self-healing, corrosion resistance, and adaptive responses, among many others. With their versatile properties, biopolymers are likely to play a crucial role in developing these intelligent coatings and adhesives. This includes the exploration of stimuli-responsive biopolymers that can dynamically adjust their properties in response to external conditions, bringing forth a new era in coating and adhesive applications with enhanced performance and sustainability (Mansouri et al., 2023; Nath et al., 2023; Rao & Wan, 2023; Sing et al., 2022).

Hybrid materials, particularly those integrating biopolymers with various molecules, have emerged as pivotal components in contemporary material science and engineering. The combination of biopolymers with other molecules offers a unique blend of properties that can be particularly seen in the development of advanced coatings and adhesives. These hybrid materials exhibit enhanced mechanical strength, flexibility, and biocompatibility, making them ideal for a myriad of applications. For instance, graphene-based hybrid coatings have garnered considerable attention due to their exceptional electrical conductivity, mechanical robustness, and thermal stability. This combination of properties makes graphene-infused coatings invaluable for applications in electronics, where they can enhance corrosion resistance and improve thermal management (Baligidad et al., 2023; Jin et al., 2023).

Beyond graphene, hybrid materials featuring biopolymers have proven instrumental in formulating environmentally friendly adhesives. By combining biodegradable polymers with functional molecules, researchers have developed adhesives that deliver robust bonding capabilities and exhibit eco-friendly characteristics. These adhesives find applications in various industries, from packaging to construction, offering a sustainable alternative to traditional adhesives composed of synthetic polymers. Many studies propose the development of hybrid materials containing biopolymers to produce interesting products, such as the development of micro-seals for dental prostheses, wood and flooring adhesives, wound healing, and many others (Kong et al., 2023; Piemjai et al., 2023; Wibowo & Park, 2023).

15.5.2 Integration of artificial intelligence in coating and adhesive development

The future of biopolymer applications is moving toward the integration of artificial intelligence (AI) in coating and adhesive development. AI algorithms, adept at analyzing extensive datasets, expedite the discovery of tailored biopolymer blends. Machine learning models enhance predictive capabilities, optimizing biopolymer selection based on specific performance criteria. This accelerates research and development, enabling the creation of customized coatings and adhesives to meet diverse industry demands efficiently. The potential impact is significant, revolutionizing the entire development lifecycle by uncovering subtle patterns, refining formulations for improved performance, and addressing specific industry challenges, such as enhanced sustainability and reduced production costs. Integrating AI in biopolymer applications is poised to offer efficient, tailored solutions that align with contemporary industry priorities.

The food industry is progressively incorporating AI into the development of food packaging (Tian et al., 2023). In a recent study, researchers explored the application of the *k*-nearest neighbor (KNN) algorithm, an AI tool proficient in analyzing small datasets. This algorithm determines the properties of new samples based on their proximity in the sample space through mathematical data analysis. The study focused on developing a fish gelatin thin film coating material that integrated essential oils to assess its impact on the physicochemical properties relevant to food packaging and preservation. The findings indicated that leveraging AI, particularly the KNN algorithm, reduces the overall cost of determining the optimal material. Through experimentation with different options like palm, clove, and oregano essential oils, the study exemplified the efficiency and cost-effectiveness of AI in enhancing the development of optimal food packaging materials (da Silva et al., 2021).

15.6 Conclusion

Functionalized biopolymer-based coatings and adhesives are emerging as promising materials for various applications in diverse industries, such as biomedical, food, textile, and environmental. By modifying the surface properties of biopolymers, functionalization techniques can enhance their performance, functionality, and sustainability, overcoming their inherent limitations and expanding their application scope. Functionalized biopolymers can create coatings and adhesives with tailored properties, such as mechanical strength, thermal stability, hydrophobicity, biodegradability, biocompatibility, bioactivity, and controlled release. These properties can contribute to developing advanced materials and technologies that address society's current and future challenges and demands. Functionalized biopolymer-based coatings and adhesives represent a valuable research area that requires further exploration and innovation, paving the way for a circular bioeconomy.

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